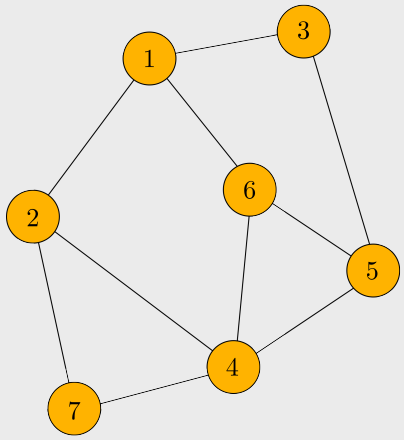
Stat 175: Homework 1

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Please submit your solutions to Gradescope by **Wednesday, February 4th, 11:59pm**.

Exercise 1 (Markov Chains). The following network represents the road network between seven different cities. There is an edge between two cities if there is a road between them. For example, there is a road between cities 1 and 2, but not between cities 1 and 5. We call i and j neighbors if there is an edge between them. We write by $\deg(j)$ the number of neighbors that the city j has. For example, $\deg(1) = 3$. A traveler is driving from town to town. At time 0, they start at the random city $X_0 \sim \pi$, where we denote by X_j the city in which the driver wakes up on the jth day. Each day the driver then drives to a new city chosen at random among all the cities that are connected to X_j. This new city is written as X_{j+1}. For example, if $X_j = 1$ then they will drive either to city 2, 3, or 6, and thus $X_{j+1} \sim \text{Unif}\{2, 3, 6\}$. We call (X_j) a *random walk* on the network.



(*a*) Prove that (X_j) is a Markov chain.

(*b*) What is the transition probability matrix of this Markov chain? Denote this transition matrix by P.

(*c*) Prove that the stationary distribution is given by $\pi(j) = \frac{\deg(j)}{20}$.

(*d*) We say that the Markov Chain is reversible if $\pi(i)P_{ij} = \pi(j)P_{ji}$ for all i, j. Prove that the Markov Chain is reversible.

(*e*) Suppose we modify the walk like so: at each step, with probability $\alpha \in (0, 1)$, the driver stays in the current city, and with probability $1 - \alpha$, they move to a uniform neighbor as before. Does this change the stationary distribution? Provide a formal proof.

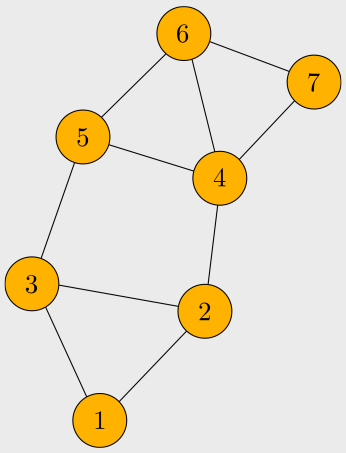
(*It might have been a while since you've read about Markov chains. For a reminder,* click here).

Exercise 2 (Linear Algebra). A matrix (A_{ij}) is called an *adjacency matrix* of a network if

(i) for all i, $A_{i,i} = 0$.

(ii) for all $i \neq j$, $A_{i,j} = \begin{cases} 1, & \text{there exists an edge between vertices } i \text{ and } j \\ 0, & \text{otherwise.} \end{cases}$

Let $D = \operatorname{diag}\big(\deg(i)\big)$ be the diagonal matrix such that $D_{i,i}$ is the degree of vertex i. The matrix $L := D - A$ is called the *Laplacian matrix* of the graph. Consider the following graph:



(a) What is the adjacency matrix of the network? What is the Laplacian matrix of the network?

(b) Compute the eigenvalues of L (feel free to use Python). Remark that the vector $\mathbf{1} = (1, \dots, 1)$ is an eigenvector.

(c) Consider a general undirected graph. Prove that for any vector $x \in \mathbb{R}^n$, the quadratic form satisfies

$$x^\intercal L x = \sum_{(i,j)\in E} (x_i - x_j)^2.$$

(d) Use this identity to prove that L is positive semi-definite and that the smallest eigenvalue is always $\lambda_1 = 0$.

(e) Let λ_2 be the second smallest eigenvalue of L. Prove that if the graph G is disconnected, then $\lambda_2 = 0$. (*Hint: Consider the vector x that is constant on one connected component and zero elsewhere*).

Exercise 3 (Statistical Inference). Let (X_i) be a sequence of i.i.d. random variables with the probability density function given by

$$f(x) := \frac{1}{2\sigma} e^{-|x|/\sigma}.$$

(a) Write out the log-likelihood function $\ell(X_1, \dots, X_n)$.

(b) What is the maximum likelihood estimator (MLE) of σ? Denote it by $\hat{\sigma}$.

(c) Is the MLE consistent? Prove your answer.

Exercise 4 (Probability Theory). Let $(U_i) \overset{\text{i.i.d}}{\sim} \text{Unif}(0,1)$ be a sequence of i.i.d. random variables and $W : [0,1]^2 \to [0,1]$ a continuous (and measurable, if you recall what that means) function. Let $(a_{i,j})$ be a sequence of Bernoulli random variables drawn in the following way:

$$a_{i,j} \sim \text{Bern}\left(W(U_i, U_j)\right).$$

Define $d_i := \sum_{j \neq i \leq n} a_{ij}$.

(a) Compute $\mathbb{E}\left[d_i \mid U_i\right]$ and $\mathbb{E}\left[d_i\right]$ for each vertex i.

(b) Compute $\text{var}\left(d_i \mid U_i\right)$ and $\text{var}(d_i)$ for each vertex i.

(c) Compute $\text{cov}(d_i, d_j)$ for each vertex pair i, j.

(d) What is the limit of $\dfrac{d_1}{n}$ and $\dfrac{d_2}{n}$? Do they converge to the same quantity? Does d_1 satisfy some form of CLT? Justify this.

(e) What is the limit of $\dfrac{\sum_{i \leq n} d_i}{n^2}$?